Stochastic Processes
Spring 2025 February 18, 2025

Homework #2: Markov Chain
Due: March 6, noon

#1 **(Running Maximum of Rolling Dice)** Let X_n be the i.i.d. outcomes of rolling dice, i.e. $X_n \in \{1, 2, 3, 4, 5, 6\}$. The running maximum is defined as $Y_n = \max_{1 \le i \le n} X_n$. Show that Y_n is a Markov chain and specify its transition probability matrix.

#2 **(Chapter 4, Exercise 3)** There are k players, with player i having value $v_i > 0$, $i = 1, ..., k$. In every period, two of the players play a game, while the other $k-2$ wait in an ordered line. The loser of a game joins the end of the line, and the winner then plays a new game against the player who is first in line. Whenever i and j play,i wins with probability $\frac{v_i}{v_i+v_j}$.

(a) Define a Markov chain that is useful in analyzing this model.

(b) How many states does the Markov chain have?

(c) Give the transition probabilities of the chain.

#3 **(Chapter 4, Exercise 5)** A Markov chain $\{X_n, n \ge 0\}$ with states 0, 1, 2 has the following transition probability matrix

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

If $P(X_0 = 0) = P(X_0 = 1) = 0.25$, find $E[X_3]$.

#4 **(Chapter 4, Exercise 14)** Specify the classes of the following Markov chains, and determine whether they are transient or recurrent:

$$P = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{2}{3} & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

#5 **(Chapter 4, Exercise 15)** Prove that if the number of states in a Markov chain is M, and if state j can be reached from state i, i.e. j is accessible from i, then it can be reached in M steps or less.

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